JSC370 Final Project Report: How Do Weather Conditions Shape Daily TTC Subway Delay Risk in Toronto?

Xiran Yin

1 Introduction

Public transit systems play a critical role in urban mobility, but reliability can be affected by environmental conditions, especially adverse weather. Weather disruptions can increase travel-time variability, reduce operating efficiency, and worsen rider experience.

In Toronto, TTC subway reliability is particularly important because the network supports very high daily ridership. This report studies whether precipitation, snowfall, temperature, and day type are systematically associated with higher daily subway delay minutes.

The analysis integrates two public datasets:

1. Open-Meteo daily weather records for Toronto.
2. TTC incident-level delay records from Toronto Open Data.

The merged panel spans 2014-01-01 to 2026-01-31 (calendar days: 4414). Because TTC record availability varies by year, modeling is conducted on days with observed incidents (2863 days), while full-calendar coverage is retained for data-accounting transparency.

The specific research questions are:

1. How strongly are precipitation and weather severity associated with daily total delay minutes?
2. Do weekday and weekend days have different delay levels once weather is accounted for?
3. Which engineered weather features are most predictive of daily delay minutes?
4. Can we accurately classify whether a day is a **severe delay day** (top quartile of delay minutes)?

The hypotheses are specified in testable null/alternative form:

1. **RQ1:** H0_1 = no association between precipitation/severity and daily delay; H1_1 = higher precipitation/severity is associated with higher daily delay.
2. **RQ2:** H0_2 = no weekday/weekend difference after weather controls; H1_2 = weekday and weekend expected delays differ.
3. **RQ3:** H0_3 = engineered weather predictors do not improve predictive fit over baseline; H1_3 = they improve predictive fit.

4. **RQ4:** H0_4 = classification does not outperform random severe-day ranking; H1_4 = at least one model yields ROC-AUC above random.

2 Methods

2.1 Model Choice and Rationale

RQ1–RQ2 are evaluated with descriptive statistics and regression coefficients, while RQ3–RQ4 use supervised prediction. The model set combines interpretable baselines with non-linear learners:

1. **Linear Regression** and **Logistic Regression** provide transparent baseline relationships for continuous and binary outcomes.
2. **Decision Tree** models allow threshold-based interactions (e.g., weather cut points) with straightforward interpretability.
3. **Random Forest** models reduce single-tree variance and generally improve out-of-sample stability.
4. **XGBoost** adds boosted non-linear capacity for potentially complex interactions between precipitation, temperature, and calendar structure.

This structure tests whether added model complexity materially improves held-out accuracy beyond linear baselines.

2.2 Data Sources

Two data streams were acquired programmatically:

1. **Weather data (Open-Meteo archive API):** Daily weather records for Toronto (requested range: 2014-01-01 to 2026-03-31), including precipitation, snowfall, temperature, and wind variables.
2. **TTC subway delay data (Toronto Open Data CKAN API):** Incident-level delay records across multiple historical CSV/XLS/XLSX resources from package `ttc-subway-delay-data`.

All source calls are made inside `scripts/final_pipeline.py`.
The weather request extends to 2026-03-31, but the merged analysis ends on 2026-01-31 because TTC incident records currently end there.

2.3 Data Wrangling and Feature Engineering

The TTC files use inconsistent column names across years. To address this, the pipeline uses conditional parsing rules:

- **Date-column detection:** candidate columns are scored by name pattern (e.g., `date`, `occ date`) and empirical parse success rate.
- **Delay-column detection:** candidate columns are scored by delay-like names (e.g., `min_delay`) and numeric-validity rate, while non-delay text fields are penalized.

- **Exception handling:** each CKAN resource is wrapped in `try/except`; failed files are logged in `outputs/ttc_resource_pull_log.csv` rather than stopping the pipeline.

After detection, records are standardized to a common incident-level schema:

- `date`
- `min_delay`
- optional `line`, `station` (when available)

Records are filtered to valid numeric delay values and aggregated to daily totals:

- `total_delay_mins` (sum of incident delay minutes)
- `total_incidents` (incident count)
- `median_incident_delay`

The final time window is defined by source overlap: weather is requested through 2026-03-31, while TTC records currently extend through 2026-01-31, so the analysis ends on 2026-01-31. A complete daily calendar is created over this overlap and merged with weather and TTC daily aggregates. Days with no TTC incident records are retained for coverage accounting and coded as:

- `total_delay_mins = 0`
- `total_incidents = 0`
- `median_incident_delay = 0`

The final daily table contains 4414 calendar days from 2014-01-01 to 2026-01-31, of which 2863 have at least one recorded incident and 1551 have zero recorded incidents. Predictive modeling uses the incident-recorded subset (2863 days) to avoid treating missing-record periods as true zero-delay operations. Coverage is uneven in a subset of years ([2017, 2018, 2019, 2020, 2021]) with a minimum annual coverage of 8.5%, which is why year-level coverage diagnostics are reported alongside modeling outputs.

Engineered predictors include:

- `mean_temp_c` (daily average of Open-Meteo `temperature_2m_max` and `temperature_2m_min`)
- `precip_intensity` (Open-Meteo `precipitation_sum / precipitation_hours` when precipitation hours > 0, else 0)
- `weather_type` (Clear / Light Rain-Snow / Heavy Rain-Snow)
- `is_weekend_int` (binary day-type feature: weekday = 0, weekend = 1)
- calendar features (`month_num`, `day_of_week`)

For model training, the numeric feature set is: `precip_mm`, `snowfall_cm`, `gust_speed`, `mean_temp_c`, `precip_intensity`, `is_weekend_int`, and `month_num`.

The three-level weather severity variable is constructed with explicit breakpoints:

- **Clear:** `precip_mm = 0` and `snowfall_cm = 0`
- **Light Rain/Snow:** `precip_mm > 0` or `snowfall_cm > 0`, but not meeting heavy criteria
- **Heavy Rain/Snow:** `precip_mm >= 10` **or** `snowfall_cm >= 5`

These thresholds operationalize weather intensity in interpretable categories for both descriptive tables and model features.

For reproducibility checks, `data/ttc_incidents_sample.csv` stores a random sample of incident records across the full multi-year window (not a first-year-only head).

The classification target is:

- `severe_day = 1` if daily delay is at or above the 75th percentile threshold (193.0 minutes) among incident-recorded days; else `severe_day = 0`.

The 75th-percentile cutoff is used as a pragmatic high-risk threshold for classification; regression results are reported in parallel to avoid over-reliance on a single binary definition of severity.

2.4 Data Exploration Tools

Exploratory diagnostics were done with Pandas summary tables (coverage/range checks), Seaborn/Matplotlib static plots (distribution and seasonal structure), and Plotly interactive visuals (time-series and weather/day-type comparisons).

2.5 Modeling Strategy

The modeling workflow maps directly to the research questions: continuous-delay prediction for RQ3 and severe-day classification for RQ4, with shared predictors and identical train/test boundaries for comparability.

2.5.1 Regression

Predicting continuous `total_delay_mins` on a 70/30 train-test split:

- Baseline: Linear Regression
- Additional tree models: Decision Tree Regressor, Random Forest Regressor
- Complex boosted model: XGBoost Regressor

Hyperparameters for tree/boosting models were set by constrained validation-oriented tuning (depth, trees, and learning-rate controls), then fixed for final held-out evaluation (e.g., RF `n_estimators=700`, `max_depth=12`; XGBoost `n_estimators=400`, `max_depth=3`, `learning_rate=0.05`).
Metrics: R^2, RMSE, and MAE on the held-out test set.

2.5.2 Classification

Predicting binary `severe_day` on a stratified 70/30 split:

- Baseline: Logistic Regression (with standardization)
- Additional tree models: Decision Tree Classifier, Random Forest Classifier
- Complex boosted model: XGBoost Classifier

Hyperparameters follow the same validation-first logic (e.g., RF `n_estimators=700`, `max_depth=10`; XGBoost `n_estimators=700`, `max_depth=4`, `learning_rate=0.03`, `eval_metric=AUC`).
Metrics: Accuracy, Precision, Recall, F1, ROC-AUC on the held-out test set.
Model interpretation includes feature-importance diagnostics. Five-fold cross-validation on incident-recorded days was also used as a robustness check: RF regression CV RMSE was 96.38 (CV R^2 0.032), and classification CV ROC-AUC was 0.584 for RF versus 0.597 for logistic regression.

Hypothesis decisions are tied to explicit rules: coefficient sign and `p < 0.05` for RQ1–RQ2, held-out RMSE gain for RQ3, and ROC-AUC above 0.5 for RQ4.

3 Results

3.1 Descriptive Patterns

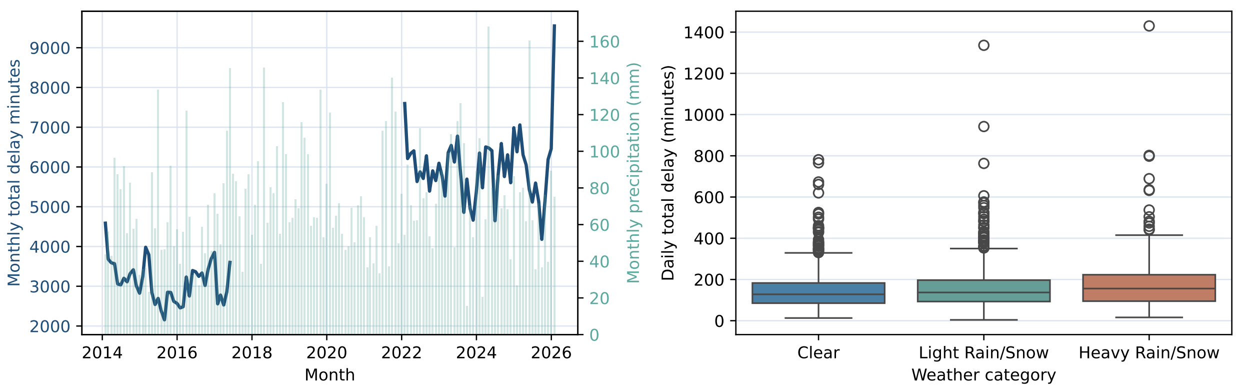
Figure 1 is a two-panel diagnostic: the left panel is the monthly trend view and the right panel is the weather-category distribution view. The left panel helps separate true temporal shifts from data-coverage gaps (notably 2017-2021), and the right panel shows a higher center and broader upper tail under heavy weather. Summary statistics are consistent with the plot patterns: heavy weather days average 189.6 minutes versus 143.7 minutes on clear days (ratio 1.32x), with positive precipitation slope (`beta = {python} round(coef_precip, 3)`, `p = {python} p_precip:.4f`).

3.2 Model Performance

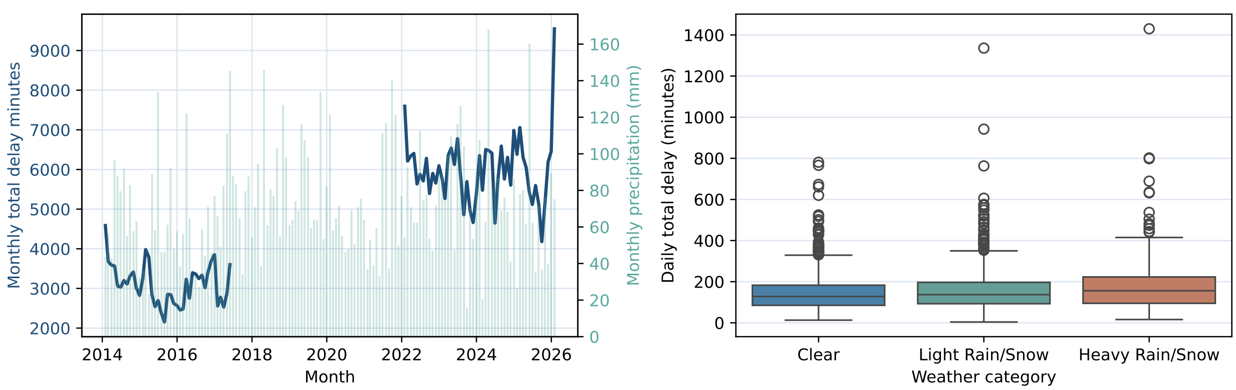
Table 1: Held-out model performance summary (RMSE/R^2 for regression; ROC-AUC/F1 for classification).

	task	model	rmse	r2	roc_auc	f1
0	Regression	Random Forest Regressor	95.5140	0.0916	NaN	NaN
1	Regression	XGBoost Regressor	95.9706	0.0829	NaN	NaN
2	Regression	Linear Regression	96.9793	0.0635	NaN	NaN
3	Regression	Decision Tree Regressor	97.3353	0.0566	NaN	NaN
4	Classification	Random Forest Classifier	NaN	NaN	0.6221	0.3613
5	Classification	Logistic Regression	NaN	NaN	0.6091	0.4185
6	Classification	XGBoost Classifier	NaN	NaN	0.5941	0.1915
7	Classification	Decision Tree Classifier	NaN	NaN	0.5301	0.2605

Table 1 shows that Random Forest Regressor has the lowest held-out RMSE (95.51) with R^2 0.092. The gain over linear regression is modest (1.51% RMSE reduction), so weather features add signal,



(a) Figure 1. Monthly trend/precipitation (left) and incident-day weather-category delay distribution (right).



(b)

Figure 1

but daily delay still has substantial unexplained variability. For classification, Random Forest Classifier has the top ROC-AUC (0.622), while logistic regression retains comparatively strong recall/F1, so model ranking is metric-specific. Feature-importance outputs are reported in the interactive-visualization artifacts and emphasize precipitation, wind, and temperature predictors.

4 Conclusions and Summary

This analysis shows that weather is a meaningful but not dominant driver of daily TTC subway delay risk. Heavier rain/snow days have higher delay levels than clear days, and weekday delay levels are materially higher than weekend levels even after adjustment. At the same time, model performance indicates that weather variables alone do not fully explain daily delay variation, so operational and network factors likely account for additional risk.

From a broader urban-infrastructure perspective, the key implication is planning prioritization: weather-aware service management can improve reliability, but practical forecasting should combine weather signals with operational indicators (e.g., incident type, line-level context, and peak-period load) for stronger day-level performance.

4.1 Hypothesis Assessment

Using the decision rules in Methods:

1. **RQ1:** reject H0_1; precipitation is positively associated with delay (`beta = {python} round(coef_precip, 3)`, `p = {python} p_precip:.4f`) and heavy-weather mean delay is 1.32x clear-weather delay.
2. **RQ2:** reject H0_2; weekend coefficient is negative and significant (`beta = {python} round(coef_weekend, 3)`, `p < 0.001`), indicating higher weekday expected delay.
3. **RQ3:** reject H0_3 with modest gain; best RMSE improves by 1.51% over linear baseline.
4. **RQ4:** reject H0_4 with moderate effect; best ROC-AUC is 0.622, above random ranking (0.5).

4.2 Limitations

1. TTC data coverage is uneven across years, which affects long-run descriptive continuity.
2. Daily aggregation masks intra-day structure (peak-hour concentration and line-specific mechanisms).
3. The severe-day definition is percentile-based (193.0 min/day) and therefore operationally useful but not a causal threshold.
4. Findings are associational/predictive, not causal.